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CARGO SHELF ASSEMBLY FOR A VEHICLE

Abstract

A cargo shelf assembly for a vehicle includes a shelf adapted to be removably engaged with the vehicle to divide at least a part of a cargo space of the vehicle. The shelf includes a plurality of retention structures to facilitate a conversion of the shelf into a table upon removal of the shelf from the vehicle. The shelf assembly also includes a plurality of latches coupled to the shelf and adapted to move between an extended position and a retracted position. In the extended position, the plurality of latches facilitates an engagement of the shelf to the vehicle. In the retracted position, the plurality of latches enables a disengagement of the shelf from side liners of the vehicle.

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Background/Summary

BACKGROUND
[0001] The disclosed subject matter relates generally to vehicles. More particularly, the disclosed subject matter relates to a removable cargo shelf for a cargo space of a vehicle.
[0002] Vehicles, such as hatchbacks, minivans, sports utility vehicles, etc., typically, include a cargo space arranged behind the rear seats that is generally visible from a rear windscreen of the vehicles. A cover assembly is generally mounted inside the vehicle to cover the cargo space from above. However, the existing cover assemblies are either fixedly mounted to the vehicle or are difficult to remove, which is undesirable.
SUMMARY
[0003] In accordance with one embodiment of the present disclosure, a cargo shelf assembly for a vehicle is disclosed. The cargo shelf assembly includes a shelf adapted to be removably engaged with the vehicle and to divide at least a part of a cargo space of the vehicle. The shelf assembly also includes a plurality of retention structures to facilitate a conversion of the shelf into a table upon removal of the shelf from the vehicle. The shelf assembly further includes a plurality of latches coupled to the shelf and adapted to move between an extended position and a retracted position. In the extended position, the plurality of latches facilitates an engagement of the shelf to the vehicle. In the retracted position, the plurality of latches enables a disengagement of the shelf from the vehicle.
[0004] In accordance with another embodiment of the present disclosure, a vehicle is provided. The vehicle includes a vehicle body defining a cargo space for storing one or more cargo and a shelf adapted to be removably engaged with the vehicle and to divide at least a part of the cargo space of the vehicle. The shelf includes a plurality of retention structures to facilitate a conversion of the shelf into a table upon removal of the shelf from the vehicle. The vehicle further includes a plurality of latches adapted to move between an extended position and a retracted position and coupled to the shelf. In the extended position, the plurality of latches engages the shelf with the vehicle body.
[0005] In accordance with yet a further embodiment of the present disclosure a shelf structure for a vehicle is disclosed. The shelf structure includes a shelf adapted to be removably engaged with the vehicle to divide at least a part of a cargo space of the vehicle and a plurality of retention structures. The shelf structure further includes a plurality of latches coupled to the shelf and adapted to move between an extended position, a retracted position, and an intermediate position disposed between the extended position and the retracted position. In the extended position, the plurality of latches facilitates an engagement of the shelf with the vehicle. In the retracted position, the plurality of latches enables a disengagement of the shelf from the vehicle. In the intermediate position, the plurality of latches facilitates a tilting of the shelf relative to a floor of the vehicle. The shelf structure further includes a plurality of legs adapted to be removably engaged with the plurality of retention structures to facilitate a conversion of the shelf into a table upon disengagement of the shelf from the vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0006] Certain embodiments of the present disclosure will be better understood from the following description taken in conjunction with the accompanying drawings in which:
[0007] FIG. **1** is a front perspective view of a vehicle, in accordance with one embodiment of the present disclosure;
[0008] FIG. **2** illustrates a rear portion of the vehicle depicting a shelf structure of a cargo shelf assembly arranged inside a cargo compartment of the vehicle with some of the components of the

vehicle removed for clarity, in accordance with one embodiment of the present disclosure;
[0009] FIG. **3** illustrates the rear portion of the vehicle depicting the shelf structure of the cargo shelf assembly arranged tilted and inside the cargo compartment of the vehicle, in accordance with one embodiment of the present disclosure;
[0010] FIG. **4** illustrates an enlarged view of a portion of the vehicle depicting a pair of notches of a side liner of the vehicle and a pair of latches of the shelf structure engaged with the pair of notches, in accordance with one embodiment of the present disclosure;
[0011] FIG. **5** is an exploded view of a latch assembly of the cargo shelf assembly, in accordance with one embodiment of the present disclosure;
[0012] FIG. **6**A illustrates a latch of the latch assembly arranged in an extended position, in accordance with one embodiment of the present disclosure;
[0013] FIG. **6**B illustrates the latch of the latch assembly of FIG. **6**A arranged in an intermediate position, in accordance with one embodiment of the present disclosure;
[0014] FIG. **6**C illustrates the latch of the latch assembly of FIG. **6**A arranged in a retracted position, in accordance with one embodiment of the present disclosure;
[0015] FIG. **7** illustrates a perspective view of the cargo shelf assembly having a plurality of legs coupled to the shelf and arranged as a table, in accordance with one embodiment of the present disclosure;
[0016] FIG. **8** illustrates an exploded view of the cargo shelf assembly of FIG. **7** depicting the plurality of legs disengaged from the shelf, in accordance with one embodiment of the present disclosure;
[0017] FIG. **9** depicts a sectional view of a portion of the cargo shelf assembly of FIG. **7** depicting a protrusion of the leg arranged inside a groove of the shelf, in accordance with one embodiment of the present disclosure;
[0018] FIG. **10** depicts a perspective view of a rear portion of the vehicle depicting the plurality of legs of stored inside a storage chamber disposed beneath a floor of the vehicle, in accordance with one embodiment of the present disclosure; and
[0019] FIG. **11** depicts a perspective view of a rear portion of the vehicle depicting the shelf structure arranged on the floor of the vehicle, in accordance with one embodiment of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0020] A few inventive aspects of the disclosed embodiments are explained in detail below with reference to the various figures. Exemplary embodiments are described to illustrate the disclosed subject matter, not to limit its scope, which is defined by the claims. Those of ordinary skill in the art will recognize a number of equivalent variations of the various features provided in the description that follows. Embodiments are hereinafter described in detail in connection with the views and examples of FIGS. **1-11**, wherein like numbers indicate the same or corresponding elements throughout the views.
[0021] FIG. **1** illustrates a perspective view of a vehicle, indicated generally at **100**, having a cargo shelf assembly **101** (shown in FIG. **2**), in accordance with one embodiment of the present disclosure. The vehicle **100** has a vehicle body **120** and includes a front end **112**, a rear end **114**, a first longitudinal side **116** (hereinafter referred to as a left side **116**), and a second longitudinal side **118** (hereinafter referred to as a right side **118**). The vehicle body **120** extends from the front end **112** to the rear end **114** and is supported on a plurality of wheels **122**. The vehicle **100** is shown as a hatchback, however, the vehicles **100** in accordance with alternative embodiments can comprise any variety of vehicles, including multi-utility vehicles, sport utility vehicles, jeeps, and trucks, etc. The exemplary cargo shelf assembly **101** may partition the vehicle interior vertically (shown in FIG. **2**.) increasing cargo management, organization, and utilization; may be easily stored flat on the vehicle floor (shown in FIG. **11**) when not in use; and may be converted into a table (shown in FIG. **7**) when removed from the vehicle.

[0022] As shown in FIGS. **2** and **3**, the vehicle body **120** defines a passenger compartment **124** and a cargo compartment **127** with the cargo shelf assembly **101** arranged inside the cargo compartment **127**. The passenger compartment **124** includes a plurality of seats **125** having one or more rear seats **126** to facilitate a seating of a plurality of passengers including a driver inside the vehicle **100**. The cargo compartment **127** is arranged at a rear of the seats **126** of the vehicle body **120** and defines a cargo space **128** to enable storage of one or more cargo inside the vehicle **100**. The vehicle body **120** also includes a pair of side liners **130***a*, **130***b* arranged inside the vehicle body **120** and extending along and mounted to longitudinal sidewalls of a frame of the vehicle body **120**. In an exemplary embodiment, the cargo shelf assembly **101** includes a shelf structure **164**, which is configured to divide and cover at least a part the cargo space **128** of the cargo compartment **127**. The shelf structure **164** is removably coupled to the side liners **130***a*, **130***b* (i.e., vehicle body **120**) and configured to move between a first position (shown in FIG. **2**) and a second position (shown in FIG. **3**) when engaged with the side liners **130***a*, **130***b*. In the first position, the shelf structure **164** is arranged substantially parallel to a floor **140** of the cargo space **128**; while in the second position, the shelf structure **164** is arranged tilted or inclined relative to the horizontally oriented floor **140**. When the shelf structure **164** is in the first position, cargo may be placed on top of the shelf structure **164**, and when the shelf structure **164** is in the second position, a person may view and access beneath the shelf structure **164**.

[0023] The side liners **130***a*, **130***b* extend in the longitudinal direction and are arranged on the opposite longitudinal sides **116**, **118** of the vehicle **100**. The side liners **130***a*, **130***b* extend from the rear end **114** of the vehicle body **120** towards the front end **112** along associated portions of the longitudinal sides **116**, **118**. It may be appreciated that the two side liners **130***a*, **130***b* may be identical in structure, construction, and assembly; therefore, for the sake of clarity and brevity, one side liner **130***a* is explained in detail. As shown in FIG. **4**, the side liner **130***a* includes a pair of notches, for example, a first notch **132** located proximate to the rear seats **126** and a second notch **134** disposed proximate to the rear end **114** of the vehicle **100**. It may be appreciated that the first notches **132** of the two side liners **130***a*, **130***b* may be arranged such that the first notches **132** are aligned with each other in the lateral direction. Similarly, the second notches **134** are aligned to each other in the lateral direction.

[0024] Moreover, the first notch **132** includes a base wall **136** disposed substantially horizontally and parallel to the floor **140** of the cargo compartment **127** and a tapered wall **142** extending obliquely upwardly from the base wall **136** and rearwardly towards the rear end **114**. Additionally, the first notch **132** includes a roof portion **144** arranged facing and opposite to the base wall **136**. The roof portion **144** is arranged spaced apart, in a vertical direction, from the base wall **136**, and therefore a chamber **148** is defined between the base wall **136** and the roof portion **144**. It may be appreciated that a width of the roof portion **144** in the lateral direction is smaller than a width of the base wall **136** in the lateral direction. An opening **150** of the first notch **132** (i.e., chamber **148**) is defined along an outer surface of the side liner **130***a*. A gap **152** may also be defined between the tapered wall **142** and the roof portion **144**. Accordingly, the first notch **132** may be formed as a depression extending in the lateral direction from the outer surface of the side liner **130***a*.

[0025] The second notch **134** may be similar to the first notch **132** except that the tapered wall **142** is omitted. Accordingly, the second notch **134** includes a base wall **154**, a roof portion **156**, a chamber **158**, and an opening **160** similar to the base wall **136**, the roof portion **144**, the chamber **148**, and the opening **150** of the first notch **132**. Alternatively, the second notch **134** may be identical to the first notch **132** and includes a tapered wall similar to the tapered wall **142** of the first notch **132**. Additionally, the side liner **130***a* includes one or more slots **162** (shown in FIG. **2**) arranged proximate to and vertically above the floor **140** of the cargo compartment **127**, and located proximate to the rear end **114** of the vehicle **100**. The slot **162** facilitates the storage of the shelf structure **164** of the cargo cover assembly **101** in a storage position on the floor **140** of the vehicle **100** (shown in FIG. **11**).

[0026] As shown in FIGS. **2**, the shelf structure **164** includes a shelf **166** that may be rectangular or square in shape and a plurality of latch assemblies **168** coupled to the shelf **166** and adapted to be engaged with the plurality of notches **132**, **134** to enable the removable engagement of the shelf structure **164** with the vehicle body **120**. The shelf structure **164** may also include a plurality of retention structures **190** on the shelf **166** and adapted to be removably engaged with a plurality of legs **300** (shown in FIGS. **7** and **8**) to facilitate a conversion of the shelf **166** into a table upon disengagement of the shelf **166** from the vehicle.
[0027] The shelf **166** extends longitudinally and laterally inside the cargo compartment **127** when engaged with the vehicle body **120** and includes a panel or body **170** having a pair of longitudinal sides **172**, **174** extending along longitudinal sides **116**, **118** of the vehicle **100** and a pair of lateral sides **176**, **178** extending in the lateral direction of the vehicle **100**. When engaged with the side liners **130***a*, **130***b*, a first lateral side **176** of the shelf **166** is arranged proximate to the rear seats **126**, while a second lateral side **178** of the shelf **166** is arranged proximate to the rear end **114** of the vehicle **100**.
[0028] The plurality of latch assemblies **168** include a pair of first latch assemblies **168***a* coupled to the longitudinal sides **172**, **174** of the shelf **166** and arranged opposite to each other. The pair of first latch assemblies **168***a* are arranged proximate to the first lateral side **176** of the shelf **166** and are arranged proximate to the rear seats **126** of the vehicle **100**. Moreover, the plurality of latch assemblies **168** includes a pair of second latch assemblies **168***b* coupled to the shelf **166** and arranged along the opposite longitudinal sides **172**, **174** of the shelf **166**. The pair of second latch assemblies **168***b* are arranged proximate to the second lateral side **178** of the shelf **166** and are arranged proximate to the rear end **114** of the vehicle **100**. It may be appreciated that the shelf structure **164** may be symmetrical about its central longitudinal axis **180** (shown in FIG. **2**) and central lateral axis **182** (shown in FIG. **3**), and the latch assemblies **168** may be identical in structure, construction and assembly with the shelf **166**; therefore, for the sake of clarity and brevity, a structure, a construction, a function, and an assembly of only one latch assembly **168** are described in detail.
[0029] As shown in FIGS. **5** and **6**A to **6**C, the latch assembly **168** includes a housing **200** removably coupled to the body **170** of the shelf **166** and having an upper member **202** and a lower member **204**, a latch **206** arranged at least partially inside the housing **200**, and a biasing member **208** engaged with the latch **206** and disposed inside the housing **200**. The upper member **202** is coupled to an upper surface **186** of the body **170** of the shelf **166**, while the lower member **204** is coupled to a bottom surface **188** of the body **170** of the shelf **166**, defining an elongated channel **210** therebetween. As shown, the elongated channel **210** extends in the lateral direction, and the latch **206** is movably arranged inside the channel **210** and configured to extend and retract between an extended position and retracted position in the lateral direction relative to the housing **200**. Moreover, the latch **206** is adapted to be displaced and held at an intermediate position between the extended position and the retracted position. Although the upper member **202** and the lower member **204** are shown and contemplated to be removably coupled to the shelf **166**, however, it may be appreciated that the upper member **202** and/or the lower member **204** may be integrally formed with the shelf **166** and may be features or components of the shelf **166**.
[0030] The latch **206** includes a rod **212** movably arranged inside the channel **210** and a handle **214** coupled to the rod **212** and extending inside a cavity **216** of the upper member **202**. The handle **214** enables a user to hold the latch **206** and to move the latch **206** between the extended position, the intermediate position, and the retracted position. In the extended position, a length of the rod **212** that extends outwardly of the housing **200**, in the lateral direction, is larger than a length of the rod **212** extending outwardly of the housing **200** in the intermediate position. In the retracted position, the rod **212** is either completely arranged inside the elongated channel **210** or a length of the rod **212** that extends outwardly of the housing **200** is smallest relative to the extension of the rod **212** outside the housing **200** in both the extended position and the intermediate position.

[0031] It may be appreciated that in the assembly of the shelf structure **164** with the vehicle body **120** and in the extended position of each of the first latch assemblies **168***a*, a front portion of the rod **212** is arranged inside the chamber **148** of the associated first notch **132** and underneath the roof portion **144**, thereby engaging the shelf structure **164** with the sideliner **130***a* or the side liner **130***b* and preventing the disengagement of the latch **206** from the side liner **130***a* or the side liner **130***b*. In the retracted position, the latch **206** is retracted inside the housing **200** such that the front portion of the rod **212** is arranged outside the first notch **132**. In the intermediate position, the front portion of the rod **212** is arranged inside the chamber **148** of the notch **132**, and away from the roof portion **144** such that the rod **212** may move out of the first notch **132** (i.e., chamber **148**) without interfering with the roof portion **144**, upon vertical movement of the shelf structure **164**. Likewise, in the assembly of the shelf structure **164** with the vehicle body **120** and in the extended position of each of the second latch assemblies **168***b*, a front portion of the rod **212** is arranged inside the chamber **158** of the associated second notch **134** and underneath the roof portion **156**, thereby engaging the shelf structure **164** with the sideliner **130***a* or the side liner **130***b* and preventing the disengagement of the latch **206** from the side liner **130***a* or the side liner **130***b*. In the retracted position, the latch **206** is retracted inside the housing **200** such that the front portion of the rod **212** is arranged outside the second notch **134**. In the intermediate position, the front portion of the rod **212** is arranged inside the chamber **158** of the notch **134**, and away from the roof portion **156** such that the rod **212** moves out of the first notch **134** (i.e., chamber **158**) without interfering with the roof portion **156**, upon vertical movement of the shelf structure **164**. Correspondingly, in the storage of the shelf structure **164** with the vehicle body **120** and in the extended position or intermediate position of at least one second latch assemble **168***b*, a front portion of the rod **212** is arranged inside the slot **162**, thereby engaging the shelf structure **164** with the sideliner **130***a* or the side liner **130***b* and preventing the disengagement of the latch **206** from the side liner **130***a* or the side liner **130***b*.
[0032] Moreover, the latch **206** may be biased to the extended position by the biasing member **208**, and is moved or displaced to the intermediate and retracted positions by moving the latch **206** against the biasing force of the biasing member **208**. Accordingly, to retain and hold the latch **206** at the intermediate position and the retracted position, the housing **200** (i.e., the upper member **202**) includes a stopper structure **220** extending inside the cavity **216** and having a first indent **222** and a second indent **224** as shown in FIGS. **6**A to **6**C. The latch **206** is engaged with the first indent **222** and the second indent **224** in the intermediate position and the retracted position, respectively. As shown, the handle **214** of the latch **206** is inserted inside the first indent **222** to retain and hold the latch **206** (i.e., rod **212**) at the intermediate position, while the handle **214** is inserted inside the second indent **224** to retain and hold the latch **206** at the retracted position. As shown, the first indent **222** and the second indent **224** are defined by the cut-outs defined in a vertically extending wall **226** of the upper member **202**. In an embodiment, the wall **226** (i.e., the stopper structure **220**) may be separate from the upper member **202** and may be integrally formed with the shelf **166**.
[0033] Referring to FIGS. **7** to **9**, the shelf **166** includes a plurality of retention structures **190**, for example, four retention structures **190**, to facilitate a conversion of the cargo shelf assembly **101** into a table. Each retention structure **190** includes a groove **192** extending inwardly from the bottom surface **188** of the body **170** of the shelf **166**. The groove **192** includes internal threads **194** (shown in FIG. **9**) to enable a removable coupling of a plurality of legs **300**, for example four legs **300**, of the cargo shelf assembly **101** with the shelf **166** to enable conversion of the shelf **166** (i.e., shelf structure **164**) into the table.
[0034] As shown, each of the plurality of legs **300** includes an elongated tube **302** having an upper end **304** and a lower end **306**, an upper cap **308** arranged at the upper end **304** of the tube **302**, and a lower cap **310** arranged at the lower end **306** of the tube **302** and removably coupled to the tube **302**. The upper cap **308** includes a body potion **312** removably engaged with the tube **302** and a protrusion **314** extending in a longitudinal direction from the body portion **312** and away from the

tube **302**. The protrusion **314** is arranged inside the groove **192**, in an assembly of the plurality of legs **300** with the shelf **166**, to convert the shelf structure **164** into a table. To facilitate the removable and secure connection of the plurality of legs **300** with the shelf **166**, the protrusion **314** may include external threads **316** (shown in FIG. **9**) adapted to engage with the internal threads **194** of the groove **192**. Although the threaded engagement of the legs **300** with the retention structures **190** is shown and contemplated, it may be appreciated the protrusion **314** may be press fitted inside the groove **192**, and in such case, the internal threads **194** of the groove **192** and the external threads **316** of the protrusion **314** may be omitted. Additionally, as shown in FIG. **10**, the vehicle **100** may include a storage chamber **400** disposed underneath the floor **140** of the cargo compartment **127** to facilitate a storage of the legs **300**.

[0035] A method for installation of the cargo shelf assembly **101** to the vehicle body **120** and use of the cargo shelf assembly **101** inside the vehicle **100** is now described. It may be appreciated that for case of description, the latches **206** associated with the first latch assemblies **168***a* are referred to as first latches **206***a*, while the latches **206** associated with the second latch assemblies **168***b* are referred to as second latches **206***b*. For installation of the shelf structure **164** inside the cargo compartment **127**, the user removes the legs **300** from the shelf structure **164**, and moves or displaces the first latches **206***a* to the extended position, as shown in FIG. **6**A. The user also moves the second latches **206***b* to the intermediate or retracted position, as shown in FIGS. **6**B and **6**C. Thereafter, the user slides the shelf structure **164** towards the rear seat **126** such that the rods **212** of the first latches **206***a* enters the chambers **148** of the first notches **132** through the gaps **152** defined between the roof portion **144** and the tapered walls **142**, and the rods **212** of the first latches **206***a* are arranged inside the chambers **148** and underneath the roof portions **144**. Thereafter, if the second latches **206***b* are in the intermediate position, the user lowers the shelf structure **164** down, such that the rods **212** of the second latches **206***b* contact the base wall **154** of the second notches **134**, and then the user inserts the rods **212** of the second latches **206***b* underneath the roof portions **156** by moving the second latches **206***b* to the extended position. If the second latches **206***b* are in the retracted position, the user lowers the shelf structure **164** down, such that rods **212** of the second latches **206***b* are placed adjacent to the base wall **154** of the second notches **134**, and the user inserts the rods **212** of the second latches **206***b* inside the chambers **158** of the second notches **134** and underneath the roof portions **156** by moving the second latches **206***b* to the extended position. In this manner, the shelf structure **164** is engaged with the side liners **130***a*, **130***b* and disposed in the first position inside the vehicle **100**, dividing and covering at least a part of the cargo space **128** from above, as shown in FIG. **2**. Also, the removed legs **300** may be stored inside the storage chamber **400**, as shown in FIG. **10**.

[0036] For moving the shelf structure **164**, and hence, the shelf **166** to a tilted position (i.e., second position, as shown in FIG. **3**), the user moves the second latches **206***b* to the intermediate position, thereby moving the rods **212** away from the underneath the roof portions **144** of the second notches **134**, while still keeping the rods **212** on the base wall **154** of the second notches **134**. Thereafter, the user tilts the shelf **166** by moving the second lateral side **178** of the shelf **166** upwardly, while keeping the first latches **206***a* arranged inside the first notches **132** and in the extended position, preventing the disengagement of the shelf structure **164** from the side liners **130***a*, **130***b* during tilting. As the second latches **206***b* are arranged in the intermediate positions, the roof portions **156** of the second notches **134** will not interfere with the rods **212** during tilting of the shelf structure **164** (i.e., shelf **166**). The user may arrange the shelf structure **164** back to the first position by moving the shelf **166** downwardly, while keeping the second latches **134** at the intermediate position. At the first position, the rods **212** of the second latches **206***b* contact the base wall **154** of the second notches **134**. Subsequently, the user moves the second latches **206***b* to the extended position to securely engage the shelf **166** (i.e., shelf structure **164**) with the side liners **130***a*, **130***b*.

[0037] For converting the cargo shelf assembly **101** into a table, the user disengages the shelf structure **164** from the side liners **130***a*, **130***b* and moves the shelf structure **164** outwardly of the

vehicle **100**. For so doing, the user moves the second latches **206***b* to either the retracted position or the intermediate position, and lifts the shelf structure **164** upwardly such that rods **212** of the second latches **206***b* come out of the second notches **134**. Subsequently, the user pulls the shelf structure **164** in rearward direction. In response to the pulling of the shelf structure **164** in the rearward direction, the rods **212** of the first latches **206***a* slides along the tapered wall **142** of the first notches **132** and comes out of the chambers **148** of the first notches **132** through the gaps **152** defined between the tapered walls **142** and the roof portions **144**, thereby facilitating disengagement of the shelf structure **164** from the side liners **130***a*, **130***b* of the vehicle **100**. Upon disengagement of the latches **206** from the side liners **130***a*, **130***b*, the shelf structure **164** and hence the shelf **166** is moved outwardly out of the vehicle **100**. In this manner, the shelf structure **164**, and hence the shelf **166**, is disengaged from the side liners **130***a*, **130***b* without operating the first latches **206***a* and keeping the first latches **206***a* to the extended position. Thereafter, the plurality of legs **300** are removed from the storage chamber **400** and are engaged with the shelf structure **164**. For so doing, the protrusion **314** of each of the plurality of legs **300** is inserted into the associated grooves **192** and each of the plurality of legs **300** is rotated in a predefined direction to enable the engagement of each of the plurality of legs **300** with the shelf structure **164**.
[0038] Further, for storing the shelf structure **164** inside the cargo compartment **127**, the shelf structure **164** and hence the shelf **166** is placed on the floor **140** of the vehicle **100**, and the second latches **206***b* are moved either to the extended positions or the intermediate positions such that the rods **212** of the second latches **206***b* extend inside the slots **162**, as shown in FIG. **11**, thereby securing the shelf structure **164**, and hence the shelf **166** on the floor **140** of the vehicle **100**, as shown in FIG. **11**.
[0039] The foregoing description of embodiments and examples has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the forms described. Numerous modifications are possible in light of the above teachings. Some of those modifications have been discussed and others will be understood by those skilled in the art. The embodiments were chosen and described in order to best illustrate certain principles and various embodiments as are suited to the particular use contemplated. The scope of the invention is, of course, not limited to the examples or embodiments set forth herein, but can be employed in any number of applications and equivalent devices by those of ordinary skill in the art. Rather it is hereby intended the scope of the invention be defined by the claims appended hereto.

Claims

1. A cargo shelf assembly for a vehicle, the cargo shelf assembly comprising: a shelf adapted to be removably engaged with the vehicle to divide at least a part of a cargo space of the vehicle; a plurality of latches coupled to the shelf and adapted to move between an extended position and a retracted position, wherein in the extended position, the plurality of latches facilitates an engagement of the shelf to the vehicle, and in the retracted position, the plurality of latches enables a disengagement of the shelf from the vehicle; and a plurality of retention structures on the shelf to facilitate a conversion of the shelf into a table upon removal of the shelf from the vehicle.
2. The cargo shelf assembly of claim 1, wherein the each of the plurality of latches is configured to move to an intermediate position disposed between the extended position and the retracted position, wherein one or more of the plurality of latches are moved to the intermediate position to tilt the shelf relative to a floor of the vehicle.
3. The cargo shelf assembly of claim 1 further comprising a plurality of legs adapted to be removably engaged with the plurality of retention structures to facilitate the conversion of the shelf into the table.
4. The cargo shelf assembly of claim 3, wherein each of the plurality of retention structures includes a groove and each of the plurality of legs includes a protrusion adapted to be inserted

inside the groove to enable the engagement of the plurality of legs with the shelf.
5. The cargo shelf assembly of claim 4, wherein each groove includes internal threads and each protrusion includes external threads adapted to removably engage with the internal threads of the groove.
6. The cargo shelf assembly of claim 1 further comprising a plurality of stopper structures to retain the plurality of latches in one or more positions.
7. The cargo shelf assembly of claim 6, wherein each of the plurality of latches is biased to the extended position, wherein each of the stopper structures includes a first indent and a second indent to retain an associated latch of the plurality of latches in an intermediate position and the retracted position, respectively.
8. A vehicle, comprising: a vehicle body defining a cargo space for storing one or more cargo; a shelf adapted to be removably engaged with the vehicle to divide at least a part of the cargo space of the vehicle and including a plurality of retention structures to facilitate a conversion of the shelf into a table upon removal of the shelf from the vehicle body; and a plurality of latches adapted to move between an extended position and a retracted position and coupled to the shelf, wherein in the extended position, the plurality of latches engages the shelf with the vehicle body.
9. The vehicle of claim 8, wherein the vehicle body includes a pair of side liners defining a plurality of notches to receive the plurality of latches arranged in the extended position.
10. The vehicle of claim 9, wherein the plurality of latches includes a pair of first latches aligned in a lateral direction, and the plurality of notches includes a pair of first notches aligned in the lateral direction and the first notches include tapered walls to enable a removal of the pair of first latches from the pair of first notches in response to sliding of the shelf in a longitudinal direction towards a rear end of the vehicle.
11. The vehicle of claim 8, wherein the vehicle body defines at least one slot arranged proximate to a floor of the vehicle to receive a latch of the plurality of latches to restrict a movement of the shelf when the shelf is stored inside a cargo compartment of the vehicle.
12. The vehicle of claim 8, wherein each of the plurality of latches is configured to move to an intermediate position disposed between the extended position and the retracted position, wherein one or more of the plurality of latches are moved to the intermediate position to tilt the shelf relative to a floor of the vehicle.
13. The vehicle of claim 8 further comprising a plurality of legs adapted to be removably engaged with the plurality of retention structures to facilitate the conversion of the shelf into the table.
14. The vehicle of claim 13, wherein each of the plurality of retention structures includes a groove and each of the plurality of legs includes a protrusion adapted to be inserted inside the groove to enable the engagement of the plurality of legs with the shelf.
15. The vehicle of claim 8, wherein the shelf further includes a plurality of stopper structures to retain the plurality of latches in one or more positions.
16. The vehicle of claim 15, wherein each of the stopper structures includes a first indent to retain an associated latch of the plurality of latches in an intermediate position and a second indent to retain the associated latch in one of the extended position or the retracted position.
17. A shelf structure for a vehicle, the shelf structure comprising: a shelf adapted to be removably engaged with the vehicle to divide at least a part of a cargo space of the vehicle; and a plurality of latches coupled to the shelf and adapted to move to an extended position, a retracted position, and an intermediate position disposed between the extended position and the retracted position, wherein in the extended position, the plurality of latches facilitates an engagement of the shelf with the vehicle, in the retracted position, the plurality of latches enables a disengagement of the shelf from the vehicle, and in the intermediate position, the plurality of latches facilitates a tilting of the shelf relative to a floor of the vehicle.
18. The shelf structure of claim 17, wherein the shelf includes a plurality of retention structures adapted to engage a plurality of legs to facilitate a conversion of the shelf into a table upon

disengagement of the shelf from the vehicle.

19. The shelf structure of claim 18, wherein each of the plurality of legs includes a protrusion and each of the plurality of retention structures includes a groove adapted to receive the protrusion to enable the engagement of the plurality of legs with the shelf.

20. The shelf structure of claim 17 further comprising a plurality of stopper structures to retain the plurality of latches in one or more positions, wherein each of the stopper structures includes a first indent to retain an associated latch of the plurality of latches in an intermediate position, and a second indent to retain the associated latch in one of the extended position or the retracted position.